

Credit Card Default Auto-ML Report

🕒 2026-05-18 04:00:48 UTC

🏆 Best: **catboost** roc_auc = 0.7790

📊 학습 24000행

테스트 6000행

Feature 19개

1 모델 비교 (holdout 기준)

model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc	best_iter (avg)
lgbm	0.7748	0.4271	3.1424	0.7917	0.7939	0.0784	0.1427	0.5468	198
xgb	0.7773	0.4233	3.1575	0.8182	0.6720	0.3474	0.4580	0.5534	179
catboost BEST	0.7790	0.4340	3.1198	0.8178	0.6590	0.3655	0.4702	0.5594	601

2 CV (OOF) 비교

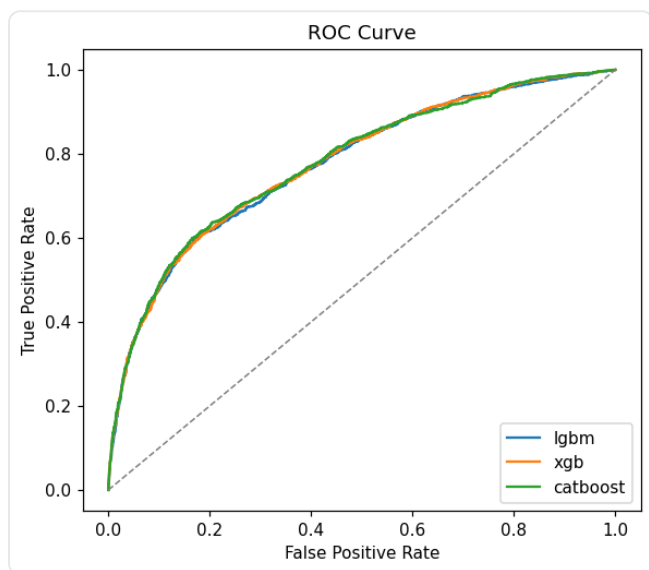
model	roc_auc	ks	lift	accuracy	precision	recall	f1	pr_auc
lgbm	0.7816	0.4327	3.1814	0.8201	0.6955	0.3321	0.4495	0.5548
xgb	0.7841	0.4349	3.1889	0.8215	0.6829	0.3601	0.4716	0.5580
catboost BEST	0.7839	0.4353	3.1946	0.8226	0.6790	0.3758	0.4838	0.5601

3 오버핏 점검 (Train vs Holdout)

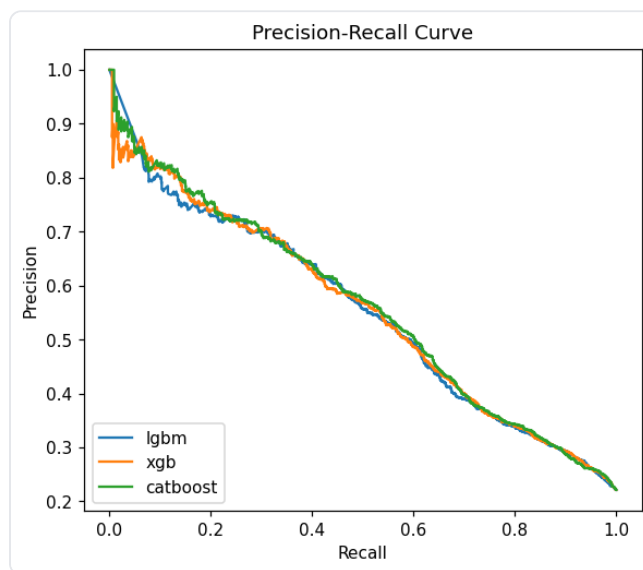
model	roc_auc			ks			lift			Train	Test
	Train	Test	Δ	Train	Test	Δ	Train	Test	Δ		
lgbm	0.8268	0.7748	+0.0520	0.5026	0.4271	+0.0754	3.2360	3.1424	+0.0936	0.7910	0.7910
xgb	0.8138	0.7773	+0.0365	0.4758	0.4233	+0.0525	3.3754	3.1575	+0.2179	0.8295	0.8188
catboost BEST	0.8077	0.7790	+0.0287	0.4624	0.4340	+0.0284	3.3490	3.1198	+0.2292	0.8287	0.8178

Δ = Train - Test. Train 은 final 모델을 학습 데이터에 fit 한 뒤 동일 학습 데이터로 in-sample 스코어링한 결과로, 의도적으로 over-optimistic 한 추정치다. Test 와의 양의 gap 이 클수록 메모리제이션(overfit) 신호이며, $|\Delta| \geq 0.05$ / $|\Delta| \geq 0.10$ 인 셀은 색으로 강조한다. (정확한 임계치는 도메인/지표에 따라 다르며 본 색은 참고용.)

ROC Curve



Precision-Recall Curve



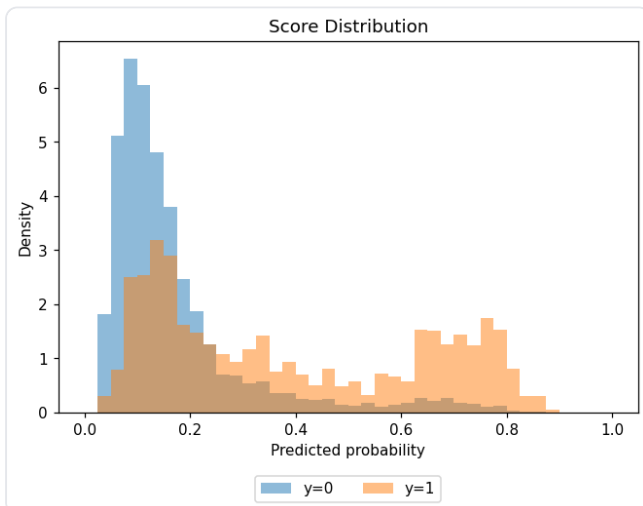
5 Best 모델 상세 — catboost

Feature Importance (relative, top 19 · 전체 분포는 feature_importance.csv)

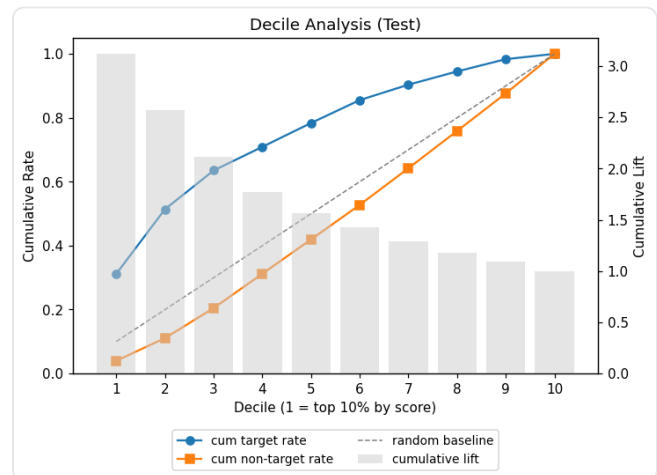
Rank	Feature	Importance	Rank	Feature	Importance
1	PAY_0	0.2524	16	PAY_AMT6	0.0263
2	LIMIT_BAL	0.0981	17	BILL_AMT6	0.0258
3	BILL_AMT1	0.0819	18	PAY_AMT5	0.0251
4	PAY_2	0.0603	19	SEX	0.0157
5	PAY_3	0.0445			
6	BILL_AMT2	0.0439			
7	PAY_AMT2	0.0437			
8	PAY_AMT3	0.0430			
9	PAY_4	0.0427			
10	PAY_AMT1	0.0412			
11	AGE	0.0359			
12	PAY_AMT4	0.0305			
13	BILL_AMT5	0.0301			
14	PAY_6	0.0297			
15	PAY_5	0.0292			

10-분위 분석 (테스트셋, 점수 내림차순) 및 점수 분포

Score Distribution by Label



Decile Analysis Chart



Decile	Score Range	Count	Target	Non-Target	Cum Target	Cum NonTarget	Cum Count	Cum %
1	0.595 ~ 0.886	600	414	186	414	186	600	10.0%
2	0.320 ~ 0.594	600	268	332	682	518	1200	20.0%
3	0.214 ~ 0.320	600	161	439	843	957	1800	30.0%
4	0.170 ~ 0.213	600	98	502	941	1459	2400	40.0%
5	0.144 ~ 0.170	600	99	501	1040	1960	3000	50.0%
6	0.122 ~ 0.144	600	95	505	1135	2465	3600	60.0%
7	0.103 ~ 0.122	600	64	536	1199	3001	4200	70.0%
8	0.086 ~ 0.103	600	55	545	1254	3546	4800	80.0%
9	0.067 ~ 0.086	600	51	549	1305	4095	5400	90.0%
10	0.026 ~ 0.067	600	22	578	1327	4673	6000	100.0%

Confusion Matrix threshold = 0.5

	Pred 0	Pred 1
True 0	4422	251
True 1	842	485

6 학습 설정 요약

설정 키	값
target_column	default
categorical_columns	SEX, EDUCATION, MARRIAGE
id_columns	client_id
preprocessing.numeric_null_strategy	median
preprocessing.categorical_null_strategy	most_frequent
preprocessing.outlier_method	percentile
preprocessing.outlier_action	clip
preprocessing.skew_method	signed_log1p
preprocessing.skew_threshold	1.0
preprocessing.scaling_method	standard
training.cv_folds	4
training.early_stopping_rounds	50
training.primary_metric	roc_auc

설정 키	값
training.random_state	42
tuning.enabled	True
tuning.n_trials	15
tuning.cv_folds	3
tuning.timeout	None

7 변수 선택 (Stability Selection)

base: lasso

부분표본 80회

임계값 0.60

채택 19 / 23

표 노출 23 / 전체 23 (전체 분포는 feature_selection.csv)

Feature	Selection Frequency	채택
LIMIT_BAL	100.0%	✓ 채택
AGE	100.0%	✓ 채택
PAY_0	100.0%	✓ 채택
PAY_2	100.0%	✓ 채택
PAY_3	100.0%	✓ 채택
PAY_4	100.0%	✓ 채택
BILL_AMT1	100.0%	✓ 채택
PAY_AMT1	100.0%	✓ 채택
PAY_AMT2	100.0%	✓ 채택
PAY_AMT3	100.0%	✓ 채택
PAY_AMT6	100.0%	✓ 채택
PAY_5	98.8%	✓ 채택
BILL_AMT5	98.8%	✓ 채택
PAY_6	97.5%	✓ 채택
PAY_AMT4	96.2%	✓ 채택
PAY_AMT5	95.0%	✓ 채택
SEX	91.2%	✓ 채택
BILL_AMT6	88.8%	✓ 채택
BILL_AMT2	71.2%	✓ 채택

Feature		Selection Frequency		채택
BILL_AMT3		56.2%		제외
BILL_AMT4		28.7%		제외
EDUCATION		21.2%		제외
MARRIAGE		15.0%		제외